

Phase 4 – Project Planning

1. Project Overview

LegalEase is planned as an AI-powered legal document generation system. The project combines a Streamlit frontend, FastAPI backend, Gemini AI integration, and document formatting modules to generate editable legal documents.

2. Project Objectives

The main objectives are:

- Generate customizable legal documents using AI.
- Collect required information such as document type, parties, terms, and effective date.
- Provide an editable preview of generated documents.
- Support TXT, DOCX, and PDF downloads.
- Provide professional formatting for generated documents.
- Handle sensitive document information securely.

3. Major Project Activities

The project activities are planned around the following stages:

1. Select the system architecture and AI model.
2. Design the Streamlit frontend.
3. Develop the FastAPI backend.
4. Integrate Gemini AI for document generation.
5. Implement document formatting and export.
6. Add preview and editing functionality.
7. Test document generation and output formats.
8. Prepare project documentation and demonstration.

4. Technology Planning

The project uses the following technologies:

- **Programming Language:** Python
- **Frontend:** Streamlit
- **Backend:** FastAPI

- **AI:** Google Gemini
- **Document Generation:** python-docx, FPDF
- **Configuration:** Python-dotenv

These technologies are used together to build and operate the LegalEase system.

/

5. Development Plan

The development of LegalEase is planned in the following sequence:

Step 1 – Frontend Development

Create the Streamlit interface for entering document type, parties, terms, and effective date.

Step 2 – Backend Development

Develop the FastAPI backend and create the /generate endpoint for processing document generation requests.

Step 3 – AI Integration

Integrate Google Gemini through the document generator module to generate legal document content from structured user inputs.

Step 4 – Document Formatting

Implement TXT, DOCX, PDF, and HTML preview functionality for the generated content.

Step 5 – Editing and Download

Provide an editable document area and download options for the generated documents.

6. Testing Plan

The system will be tested using different document inputs and generated outputs.

Testing includes:

- Checking document generation with sample inputs.
- Verifying the generated document preview.
- Testing document editing.

- Testing TXT, DOCX, and PDF downloads.
- Checking formatting, logo, footer, headings, bullets, and terms table.

7. Deployment Plan

The project will use a Python virtual environment with the required dependencies and configuration through a .env file.

The FastAPI backend can be launched using Uvicorn, while the Streamlit application is launched separately for the user interface.

8. Project Deliverables

The planned project deliverables include:

- Working LegalEase application.
- AI-generated legal documents.
- Editable document preview.
- TXT, DOCX, and PDF output files.
- Project documentation.
- Project demonstration video.
- Public GitHub repository containing the project files.

9. Expected Outcome

The planned outcome is a functional LegalEase system that allows users to enter required document information, generate legal content using Gemini AI, preview and edit the generated document, and download it in different formats.

10. Project Completion Plan

The project will be completed through the following sequence:

Planning → Development → AI Integration → Testing → Documentation → Demonstration

Each stage contributes to the development and final presentation of the LegalEase application.

11. Conclusion

The Project Planning phase organizes the major development, testing, deployment, documentation, and demonstration activities required for LegalEase. The plan provides a structured approach for developing the application from initial setup to final project demonstration.